



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

CSE 1110: Introduction to Computer Systems - Final Examination (F)

Marks: 20

SET A

Answer all the questions

1

Suppose you're creating a Course Enrollment Fee System. Students from different departments pay different fees per credit. Some discounts apply for high credit loads.

Department	Fee Per Credit	Discount
CSE (C)	1800	10% if credit taken ≥ 15
BBA (B)	1600	8% if credit taken ≥ 18
EEE (E)	1400	12% if credit taken ≥ 20

If total fee is greater than 25,000 then 5% VAT will be added after discount.

Write a program that takes a character input for the department (C, B or E) and the total credits taken by a student. Calculate the total fee the student has to pay. Print the Base Fee (Total Credit Taken x Fee Per Credit), amount after discount, Amount of VAT, and total fee.

Sample Input	Sample Output
Enter Department: B Total Credit Taken: 20	Base Fee: 32000 Tk After Discount: 29440 Tk VAT (5%): 1472 Tk Final Fee: 30912 Tk
Enter Department: C Total Credit Taken: 10	Base Fee: 18000 Tk After Discount: 18000 Tk VAT (5%): 0 Tk Final Fee: 18000 Tk

2

A parking gate generates numeric passcodes for each car. Cars with even digit passcode are allowed to pass through the gate. Write a program that takes a positive integer passcode and counts how many digits it has. If the number of digits is even, print "Allowed to Pass", otherwise, print a message "Not an even digit passcode".

Sample Input	Sample Output
Enter Passcode: 9425	Number of Digits: 4 Allowed to Pass
Enter Passcode: 147	Number of Digits: 3 Not an even digit passcode